emergency transport, and inter-business district mobility. While widespread adoption will require advancements in vehicle certification, vertiport infrastructure, airspace integration, and public trust, the rise of passenger drones signals the beginning of a multi-layered urban mobility ecosystem, where aerial transportation becomes as accessible and routine as ground-based travel.

9.4.2 Infrastructure challenges for integrating drones into cities

The successful integration of passenger drones into urban environments requires the development of specialized infrastructure capable of supporting safe, efficient, and coordinated aerial operations. Central to this is the establishment of vertiports designated take-off and landing facilities strategically located on rooftops, transit stations, commercial centers, and peripheral transport hubs. These vertiports must provide not only boarding and disembarkation platforms, but also charging stations, maintenance bays, passenger handling areas, and digital communication interfaces that connect aircraft to traffic management networks. Designing these facilities in dense cities presents spatial, architectural, and zoning challenges, particularly where building height regulations and land-use constraints are tightly controlled.

In addition to physical infrastructure, aerial traffic management is a critical requirement for large-scale urban drone operations. Cities will need unmanned traffic management (UTM) systems capable of monitoring flight paths, regulating airspace corridors, preventing collisions, and coordinating interactions between drones, traditional aircraft, and ground-based aviation services. This requires real-time data exchange supported by high-speed communication networks, including 5G, satellite connectivity, and low-latency ground relay systems. Furthermore, operating in urban landscapes introduces additional risks such as signal interference from tall buildings,